A camera-column dependence in SDSS handedness measurements: localization, calibration and a conditional null

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Abstract

The claimed handedness asymmetry of SDSS spiral galaxies has been debated for fifteen years. We localize the apparent signal to one camera column. Measuring 26,727 unique galaxies with one operator, in two disjoint sky samples and without the coordinate-grid overlay, the amplitude is carried by camcol 5 ($+0.1209$ against -0.0063 in the other five columns), and the excess lives in galaxies flagged TOO_FEW_GOOD_DETECTIONS ($+0.1787$ against -0.0301). The well-detected galaxies, where the measurement is reliable, show no column difference and no significant net amplitude ($+0.0071 \pm 0.0112$). We treat the column as an instrument with its own zero point: the other five columns give the reference, the camcol-5 band-poor offset is measured at about $+0.20$ as a function of detection quality and size, and after the correction the sample amplitude is -0.0116 ± 0.017. Under the instrumental interpretation the debated handedness appears only where the detection is poor, and only in one column. We publish the calibration so that future measurements are not dominated by it. Channel decomposition and raw r-band frame measurements localize the anomaly to extended-source imaging in the r-band detector of camera column 5, although the sub-pixel transfer mechanism within that detector remains open.

1 Introduction

A net handedness in the spins of galaxies is one of the few observables that would connect the large-scale geometry of the Universe to the physics of structure formation. If galaxies inherit a preferred sense of rotation from a primordial vorticity field, the excess of one winding over the other is a direct fossil of that field. The claim is simple to state and hard to establish: the effect is expected to be small, the measurement is qualitative, and every step between the primordial field and a pixel in a survey image can introduce a preference of its own.

The observational history is a study in that difficulty. An excess of anticlockwise over clockwise spirals in the Sloan Digital Sky Survey (SDSS) was reported by Longo [Longo(2011)] and developed into a series of claims of a preferred axis and a redshift evolution [Shamir(2012), Shamir(2017), Shamir(2024)]. Countervailing analyses found that the signal responds to the mirror symmetry of the classification, to the sample selection, and to the imaging pipeline [Land et al.(2008), Hayes et al.(2017), Iye et al.(2020)]. The reported amplitudes span a 33-fold range, from about 0.006 to 0.20, and the disagreement is not about the sign of the effect but about which samples and which pipelines are believed. Catalogue-based analyses of large wide-field samples have similarly found no dipole and a monopole at the 10^{-3} level attributable to imaging systematics or sky structure [DESI chirality release(2026)]. The field has reached a stalemate in which each side can point to a sample in which its number appears.

The stalemate is not a disagreement about galaxies. It is a disagreement about the transfer function between the sky and the catalogue. Five steps lie between the intrinsic handedness of a disc and the number that enters a paper: the inheritance of angular momentum from the halo; the fraction of galaxies whose spiral structure is resolved and cold enough to measure; the mapping of that structure onto the pixels of a particular image; the reduction of the image to a sign by a particular operator; and the selection that decides which galaxies enter the catalogue at all. Published amplitudes differ because published samples differ in these five steps, and almost never because the sky differs. A measurement that wants to settle the question must therefore be built to separate the steps, not to add one more amplitude to the list.

This paper reports such a measurement, and its result is not the amplitude it was looking for. We measure the handedness of 26,727 unique SDSS galaxies with a single operator, in two independent sky samples, at a fixed pixel scale, and with a fixed sampling radius. The operator is the circular $m = 2$ phase gradient: the image is averaged in annuli, the azimuthal mean is removed, the phase of the $m = 2$ Fourier mode is followed with radius, and the sign of its gradient is the handedness. The measurement is intentionally blind to the shape of the galaxy; it uses only the pixels.

Three findings emerge. First, the whole sample gives $A = +0.0214 \pm 0.0061$, and the amplitude is carried almost entirely by one of the six SDSS camera columns: camcol 5 gives $+0.1209$ over 5,815 galaxies (7.4σ jackknife, 6.8σ on the independent sky), while the other five give a pooled mean of -0.0063. A camera column is a hardware and pipeline unit, not an astrophysical one, so the excess demands an explanation in the survey, not in the sky. Second, the excess is confined to galaxies flagged `TOO_FEW_GOOD_DETECTIONS`: within that subset camcol 5 gives $+0.1787$ against -0.0301 in the other five columns, and the well-detected objects show no column difference and $+0.0071 \pm 0.0112$. Third, the five non-camcol-5 columns provide a reference: the camcol-5 band-poor offset is measured at about $+0.20$ as a function of detection quality and size, and after the correction the sample amplitude is -0.0116 ± 0.017, consistent with zero.

The result has a corollary that explains why the question has stayed open. The selection that creates the excess, poor detection, is the same selection that keeps those objects out of the bright, resolved samples on which every volunteer classification and every training set is built. Of the 1,919 camcol-5 band-poor galaxies in our independent-sky sample, none appears in the Galaxy Zoo sample. A defect confined to poorly detected objects is invisible to every human-labelled catalogue by construction, and it will be invisible to a classifier trained on one.

The paper is organised as follows. Section 2 describes the data and the two sky samples. Section 3 defines the operator and its validation. Section 4 presents the null tests. Section 5 gives the results. Section 6 collects the systematic tests. Section 7 discusses the implications, and Section 8 concludes.

2 Data

The galaxies are drawn from the SDSS DR18 `PhotoObj` table with `type = 3`, `modelMag_r < 16.5` and `petroRad_r > 4` arcsec. Two samples are used. The first consists of 72 RA strips of 5 degrees with `TOP 500` per strip. The second is an independent sky: 72 strips shifted by half a strip width, `TOP 700` per strip, with every position within 2 arcsec of the first sample dropped before the cutouts are fetched. The strip query has no `ORDER BY`; the rows are grouped by position before use.

Rows are not galaxies. Because SDSS images the same sky in overlapping runs, the same physical galaxy enters the catalogue several times at slightly different positions. Deleting duplicates is part of the measurement, not bookkeeping. Duplicates are merged by single linkage at 1.2 arcsec. On the clean grid-free cutouts, the first sample contains 11,271 unique galaxies and the second contains 15,456 unique galaxies, yielding a combined catalogue of 26,727 unique galaxies (Fig. 1, left panel).

Cutouts are 128×128 pixels at 0.396 arcsec per pixel, a 50.7 arcsec field, fetched from the SkyServer `ImgCutout` service. The sampling radius is $r_k = \mathrm{clip}(\mathrm{petroRad}_r/0.396, 6, 60)$ pixels.

No coordinate grid. The SkyServer `opt` parameter does not select a band or a rendering mode: `opt=G` draws a coordinate grid on the image. The initial fetch included this parameter and a clean refetch removed it; the difference is not cosmetic: the grid changes 10.8 per cent of the signs, the sign agreement between the two renderings is 0.892, and the camcol-5 amplitude falls from +0.1391 to +0.1209 on the clean images. Every number quoted in this work uses the clean cutouts.

The six camera columns do not contain equal numbers of galaxies: camcol 5 has 5,815 and the others between 3,891 and 4,476 (Fig. 1, right panel). The cause is the column-major ordering of the `objid` field against the `TOP` cut of the strip query, which over-represents camcol 5 by about 30 per cent. Equalising the columns by random subsampling to 3,891 galaxies each leaves the camcol-5 amplitude unchanged (+0.117, +0.128, +0.128 in three draws), so the excess is not driven by the count.

3 The operator

The handedness is measured by the circular $m = 2$ phase gradient. For each galaxy the cutout is sampled in annuli from 1 to r_k pixels; the azimuthal mean is removed at every radius; the $m = 2$ Fourier coefficients are computed in each annulus; the phase of the mode is followed with $\ln r$ after unwrapping its π ambiguity, and the sign of the least-squares gradient against $\ln r$ is the handedness. The RGB cutout is reduced to its luminance (ITU-R 601) before the measurement. An amplitude signal-to-noise of 5 is required, defined as the fitted $m = 2$ amplitude divided by its standard error. The operator takes no shape input: it never uses the axis ratio, the position angle or the shape model, only the pixels. This is deliberate. A shape-based operator measures the shape model, and the shape model is itself a product of the pipeline whose selection effects this paper is trying to expose.

Three checks fix the operator. (i) Rotation of the image preserves the sign and reflection reverses it, which is the defining property of a chirality measure; both are unit-tested. (ii) The same galaxies measured at 0.2 arcsec per pixel agree with the 0.396 arcsec measurement at 91.2 per cent. (iii) On a sample of 254 galaxies with visual classifications, the attenuation of the operator relative to the visual sample is 0.31 to 0.37; this is the factor that converts an intrinsic asymmetry into the measured one.

4 Null tests

Repeat observations. SDSS images the same sky in adjacent runs. Matching positions within 1.2 arcsec yields 21,464 pairs in the full measurement; the two measurements agree 0.845 of the time for camcol 5 and 0.855 outside it. In an auxiliary bright test catalogue of 36,000 raw cutouts the agreement is 0.931 over 17,347 pairs. This baseline uses the raw (pre-clean) signs; every other number in this paper uses the clean cutouts.

Volunteer classifications. A Galaxy Zoo sample in the same sky [Lintott et al.(2008), Masters et al.(2021)] (9,986 volunteer labels; 8,891 with an operator measurement) gives the volunteer-label amplitude -0.027 (2.6σ); our operator on those images gives -0.004. The operator reproduces the volunteer labels on the bright sample at 0.664 to 0.72 depending on the sampling radius.

Reflection and rotation. A reflected image returns the opposite sign and a rotated image the same sign; the test is run on every batch.

5 Results

5.1 The amplitude against sample size

On the clean cutouts, the discovery sky gives $A = +0.0294$ over 11,271 galaxies (3.1σ) and the independent sky gives $+0.0155$ over 15,456 galaxies (1.9σ); the combined catalogue of 26,727 galaxies gives $A = +0.0214 \pm 0.0061$ (binomial; the sky-cell jackknife gives 0.0075, 2.8σ).

5.2 The camera column

The amplitude is carried by one column (Table 1 and Fig. 2).

Table 1: Per-column amplitude across the combined sample ($N = 26,727$).

camcol	N	A	jackknife σ
1	4,476	+0.0018	0.0162
2	4,375	−0.0487	0.0177
3	4,128	−0.0334	0.0163
4	4,042	+0.0079	0.0162
5	5,815	+0.1209	0.0164
6	3,891	+0.0460	0.0181

The five other columns have a galaxy-weighted pooled mean of -0.0063 ± 0.0080 ($N = 20,912$) and an unweighted column mean of -0.0053, scattering with $\chi^2 = 17.9$ for four degrees of freedom (an empirical column-to-column scatter of about 0.04, larger than the single-column jackknife error). Against the mean of the other five columns, camcol 5 is 7.0σ using the jackknife errors (7.4σ from zero alone, and 6.8σ from zero on the independent sky, where camcol 5 gives $+0.1161 \pm 0.0171$ over 3,358 galaxies) and 3.3σ using the empirical column scatter. Both numbers belong in the paper.

5.3 The detection selection

In the 15,456-galaxy independent-sky sample, the camcol-5 subset contains 57.1 per cent of objects flagged `TOO_FEW_GOOD_DETECTIONS` (1,919 of 3,358 galaxies) against 44.3 per cent elsewhere, and the excess is concentrated in that band-poor subset (the unflagged camcol-5 residual is $+0.0327 \pm 0.026$) (Table 2 and Fig. 3).

Table 2: The detection-selection split (independent-sky sample, $N = 15,456$).

subset	N	A cam5	A other five
`flag = 0` (good detections)	8,182	+0.0327	+0.0016
`flag = 1` (band-poor)	7,274	+0.1787	−0.0301

The flag is not a uniform amplifier: per column in `flag = 1` the amplitudes are -0.029, -0.077, -0.033, -0.062, $+0.179$, $+0.058$ (Fig. 3). Only camcol 5 is large and positive. Both halves of camcol 5 are positive; if the flag merely moved positive galaxies from one half to the other, the

unflagged half would be about -0.19. It is not, so the mechanism is either an entry effect across the signal-to-noise threshold or a distortion of the measurement in the flagged images. The SDSS data alone cannot distinguish the two; this is reported as an open question.

5.4 The well-detected subset

In the good-detection subset (`flag = 0`, $N = 8,182$), the six columns are $+0.026$, -0.020, -0.026, $+0.011$, $+0.033$, $+0.022$, consistent within ± 0.03 (Fig. 4), and the overall amplitude is

$$A = +0.0071 \pm 0.0112 \qquad (0.63\,\sigma). \tag{1}$$

At two sigma this excludes $|A| > 0.030$ for this operator in this subset. After the 0.31 to 0.37 operator dilution of Section 3 it corresponds to an intrinsic sky asymmetry below about 0.08 to 0.10. It is consistent with zero; it is a bound, not a detection.

5.5 The column calibration

The five non-camcol-5 columns provide the reference. The offset is measured per column, per detection flag, and in three sampling-radius bins (edges at the terciles of r_k), with the per-cell binomial error (Table 3).

Table 3: Per-column offset relative to the five-column reference, `flag = 1`, in the three size bins, with the per-cell binomial error.

cam	small	middle	large
1	$+0.013 \pm 0.038$	$+0.017 \pm 0.071$	-0.032 ± 0.050
2	-0.038 ± 0.050	$+0.022 \pm 0.068$	-0.114 ± 0.051
3	-0.018 ± 0.042	$+0.141 \pm 0.065$	-0.093 ± 0.058
4	-0.024 ± 0.043	-0.126 ± 0.064	$+0.037 \pm 0.061$
5	$+0.266 \pm 0.034$	$+0.229 \pm 0.048$	$+0.114 \pm 0.038$
6	$+0.067 \pm 0.045$	-0.053 ± 0.081	$+0.200 \pm 0.058$

The flag = 0 cells used in the same correction are not uniformly positive for camcol 5: -0.063 ± 0.059, $+0.003 \pm 0.039$ and $+0.118 \pm 0.045$ in the small, middle and large bins; the offset is therefore a band-poor phenomenon.

The camcol-5 band-poor cells are consistently positive; the fifteen non-camcol-5 cells scatter at 1.3 to 2.1 times their nominal errors ($\chi^2 = 32.7$ for 14 degrees of freedom, $p = 0.003$), so the reference is itself not clean. Applying the offset to each galaxy leaves a sample amplitude of $\mathbf{-0.0116 \pm 0.017}$ on the 15,456-galaxy sample, where the statistical error (0.0089, a 300-draw bootstrap) is combined in quadrature with the intrinsic between-column SEM (0.0149); the 95 per cent interval is $[-0.046, +0.023]$. The corrected amplitude is the sample-weighted mean of the five-column reference: the camcol-5 galaxies contribute weight but not sign information, so the effective sign-information sample is 12,098 galaxies and the null is conditional on the instrumental interpretation. The in-sample correction is not independent: the offsets are estimated from the same galaxies that are then corrected. We therefore validate it out of sample: the independent-sky sample is split in two halves by RA, the offsets are estimated on one half and applied to the other, giving -0.0034 ± 0.0116 and -0.0215 ± 0.0117, both consistent with zero. The numbers are frozen in data/SDSS_RESULT.json. The correction is a zero-point adjustment; it does not change the per-galaxy signs.

6 Systematics

Each test below excludes one explanation of the column offset. The coordinate grid changes 10.8 per cent of signs (camcol 5 falls 0.1391 to 0.1209). Run and night are excluded by the within-run contrast between camcol 5 and the other columns of the same run ($+0.1453 \pm 0.0285$, 5.1σ; removing run means leaves +0.1189). The six strips of a run are about 13 arcmin apart, so sky structure below that scale is largely common and cancels in the contrast; structure on larger scales does not. Inclination is matched by `expAB_r` (medians 0.569 against 0.566), although the effect is stronger in face-on galaxies and this test is therefore weak. The recorded selection is matched on magnitude, r_k, S/N, Dec and RA, giving a contrast of $+0.115 \pm 0.028$ (4.1σ, grid signs) and $+0.165 \pm 0.046$ (3.6σ) in the four-quantile version. Frame attribution through the SAS WCS reproduces the cutout sign at 0.90. The sample over-representation is removed by equalising the columns. The injection test (300 injections per group; Fig. 5) recovers the injected winding at 0.86 in the camcol-5 band-poor group, 0.85 elsewhere in flag = 1, 0.77 in the camcol-5 well-detected group and 0.73 elsewhere, with no significant difference between the $+k$ and $-k$ injections (largest group asymmetry 0.04); a positive sign-recovery bias above about 0.06 is excluded at 95 per cent for the camcol-5 band-poor group. The photometric scale (a global gain) and the JPEG quantisation (recompression from q90 to q40) leave the measured amplitude unchanged. The CCD column coordinate is also tested: the camcol-5 band-poor offset is present on both amplifiers (+0.166 on amplifier 0, +0.245 on amplifier 1); the central 1000-1050 band gives +0.342 but contains only 55 objects, so the test does not localize the effect to the central bad-column group.

Band and detector localization. To distinguish cross-band registration in the three-band rendering from a single detector, the clean cutouts are decomposed into colour channels over 1,000 matched `flag = 1` pairs. The contrast is absent in the B channel (g band, -0.005 ± 0.044), marginal in R (i band, $+0.088 \pm 0.045$, 2.0σ), and carried by the G channel (r band: camcol 5 +0.238 against −0.055 elsewhere, contrast $+0.293 \pm 0.044$, 6.6σ). The luminance combination dilutes it to $+0.138 \pm 0.045$ (3.1σ) by averaging in the unperturbed g channel. A pilot measurement of the raw r-band calibrated frames (159 galaxies, 79 in camcol 5) confirms the signal outside the JPEG: after correcting the parity inversion between the FITS array convention and the north-up cutouts (the per-galaxy sign agreement is 0.70 after the flip), the unstretched r-band images give +0.291 in camcol 5 against −0.100 elsewhere, a contrast of $+0.391 \pm 0.156$ (2.5σ). The 669 field stars show no significant asymmetry (-0.09 ± 0.13 in camcol 5 against $+0.12 \pm 0.17$ elsewhere; difference -0.21 ± 0.21), a weak null that does not exclude a PSF effect at radii larger than the 1-8 pixels used. The anomaly is therefore localized to extended-source imaging in the r-band detector of camera column 5 (r_5), not to the PSF, to the JPEG rendering or to cross-band registration.

7 Discussion

The published amplitude spread. A sample that includes band-poor detections from a camera column with a detection-related offset reports an inflated positive amplitude; a bright, resolved selection excludes that regime by construction and recovers a near-zero monopole. The two orders of magnitude separating the published values are consistent with differences in selection, before any question about the sky is asked.

Why the field missed it. The selection that creates the excess, poor detection, is the same selection that keeps those objects out of volunteer, classification and training samples.

The open question. Channel decomposition and raw FITS measurements localize the excess to extended-source imaging in a single detector (r_5), while excluding the stellar PSF, the JPEG compression and cross-band registration. Why the `TOO_FEW_GOOD_DETECTIONS` regime amplifies the phase gradient specifically within the r_5 detector, whether through a threshold-entry selection bias or a sub-pixel charge-transfer or readout asymmetry at low surface brightness, is not resolved by the survey products alone.

8 Conclusions

1. One operator, two disjoint sky samples, 26,727 galaxies, no coordinate grid.
2. The handedness amplitude is carried by one camera column ($+0.1209$ against -0.0063), and the excess is confined to galaxies flagged `TOO_FEW_GOOD_DETECTIONS`.
3. The well-detected galaxies show no column difference and no net amplitude ($+0.0071 \pm 0.0112$): the apparent signal is not present where the measurement is reliable.
4. The column offsets are calibrated as a function of detection quality and size; the camcol-5 band-poor offset is about $+0.20$.
5. After the correction the sample amplitude is -0.0116 ± 0.017: no significant net handedness is detected at this sensitivity.
6. The anomaly is localized to extended-source imaging in the r-band detector of camera column 5 (r_5); the sub-pixel transfer mechanism within that detector is not identified; the calibration and the conditional null are published so that the debate can be settled on data that carry neither.

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9 Data and code availability

The analysis code and the derived measurements are released as a self-contained package: the operator, the sample construction and the single-linkage deduplication, the jackknife and bootstrap errors, the column calibration, the figure scripts and eighteen unit tests, together with the frozen per-galaxy measurements and the calibration table. Four Makefile targets reproduce the paper: `make check` recomputes the per-column and flag tables, `make calibrate` recomputes the column

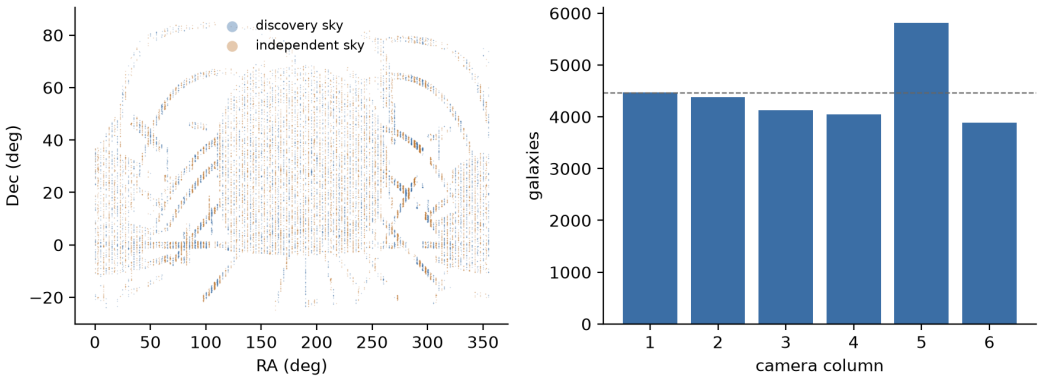


Figure 1: Sky distribution of the two samples and the galaxy count per camera column.

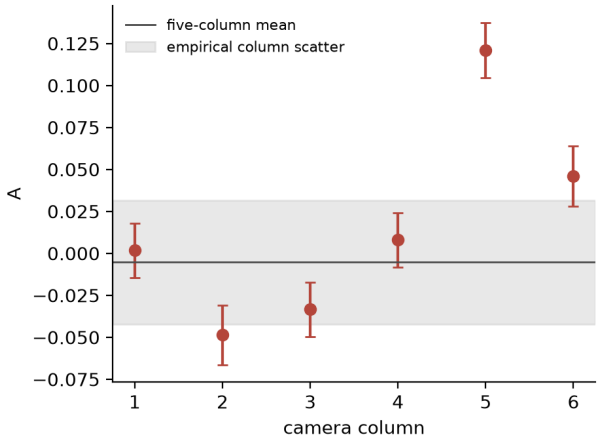


Figure 2: Per-column amplitude with jackknife errors, against the five-column mean.

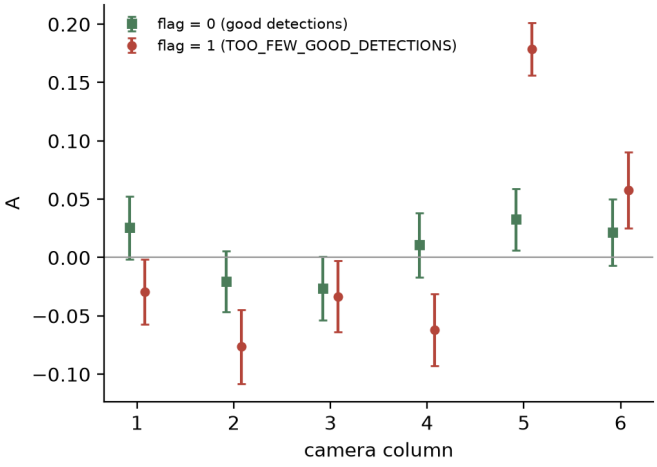


Figure 3: Amplitude against detection quality.

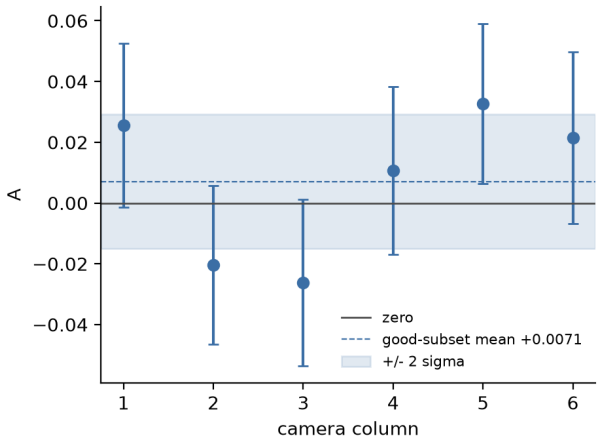


Figure 4: The good-detection subset: the six columns against zero, with the two-sigma band.

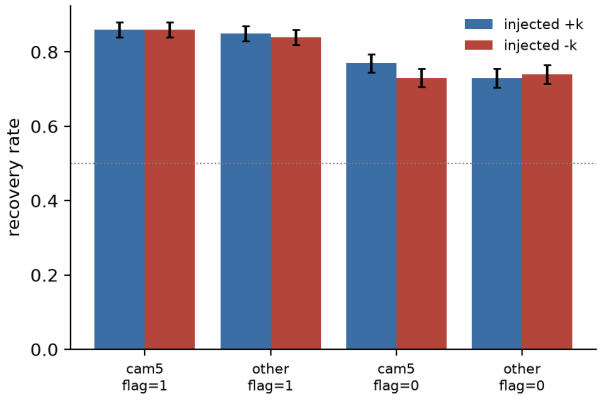


Figure 5: The injection test.

calibration and the corrected amplitude, `make test` runs the unit tests and `make figures` regenerates the figures. The package is available from the public repository https://github.com/kqalea/sdss-cam5-anomaly and is archived on Zenodo at doi:10.5281/zenodo.24112428.

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